MIT EECS

Electrical Engineering | Computer Science | Artificial Intelligence + Decision-making

Applying to Graduate School

Commitment to Building Community

Our Promise

- We ask all EECS community members to make the following promise to each other: We commit, as members of the EECS community at MIT, to take an ***active role*** in making our department welcoming and inclusive to everyone. To do that, we pledge to embrace the following values:
  - To treat all members of our community with respect.
  - To refrain from making assumptions or judgments about my fellow community members.
  - To embody empathy and compassion in all my personal interactions and communications.
  - To routinely reach out to others to offer assistance and support.
  - To engage in respectful discourse with members whose perspectives differ from my own.
  - To continually reflect on both the *intended and unintended* consequences of my comments and actions.
- We commit to living out these values because we understand that our department is stronger and more effective in its mission to change the world when we are united as a community. We understand, and agree, that belonging and inclusion must be strongly encouraged, viewed as an asset and a strength, worked toward with dedication and perseverance, and rewarded.

THE EECS Department at MIT

EECS is currently organized by three overlapping faculties (or entities) comprised of electrical engineering (EE), computer science (CS) and artificial intelligence and decision-making (AID). The EECS faculty align their research and teaching activities with either one, two or three of the aforementioned faculties; today, the number of EECS faculty is near 160. The doctoral (PhD) program is our premier graduate degree. Although our PhD students will earn a master's degree as part of their PhD degree, EECS at MIT does not have a terminal Master of Science degree. Presently, our PhD graduate student population has over 1000 active students, with 385 woman identifying students (33% overall); of the graduate student body in EECS 42% have an international citizenship.

Research guidance and mentorship is provided by 136 EECS faculty, and 83 faculty across MIT and also by faculty at nearby peer universities. Our graduate student body is highly accomplished, receiving a wide assortment of fellowship awards; roughly one-third of our current students are supported by fellowships, training grants and internships. The remainder of our PhD grad students are supported by either a research assistantship or by a teaching assistantship. With satisfactory academic progress, all of our graduate students are fully supported throughout the duration of their graduate studies.

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Why I Chose MIT

A graduate of the Computer Science program, Ariel Anders is from Bakersfield, California. "I chose MIT because I wanted to become a researcher and educator. MIT has the most robotics labs doing work I'm interested in. The MIT EECS graduation requirements are research oriented- there aren't frivolous hoops to jump through like at other schools. MIT is at the frontier of education with its online courses and offers some of the best engineering labs in the world." Dr. Ariel Anders '2019 is now a roboticist at Robust.AI.

Saeed Mehraban is a graduate of the Electrical Engineering program from Shiraz, Iran. "Perhaps I can't say why I chose MIT, but I can tell you why I will choose MIT again. MIT made me aware of real-world problems, gave me the insight to discern central questions and sufficient courage to tackle them- no matter how hard they are." Dr. Saeed Mehraban '2019 is Assistant Professor of CS at Tufts University.

Why I Chose MIT

Vikas Garg is originally from Meham (Rohtak, India) and is a graduate of the Computer Science program. "MIT inspires you to be the best by learning from the best – I chose MIT because there is no other place that **M**akes **I**ntellect **T**oil!" Dr. Vikas Garg '2020 is co-founder and chief scientist of Yai Yai Ltd and a CS faculty member at Aalto University, Finland.

Mandy Korpusik is from Campbell, California and is a graduate of the Computer Science program. "When I came to the visit weekend, I fell in love with the warm, welcoming EECS community here at MIT, especially its support for graduate women, and I became interested in joining the Spoken Language Systems group after learning about all the exciting projects everyone was working on." Dr. Mandy Korpusik '2019 is Assistant Professor of CS at Loyola Marymount College.

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Contents of Booklet

- ❖ Applying to Grad School
  - ❖ What Does the Application Contain?
  - ❖ Criteria used by admissions committees?
- ❖ Elements of EECS Graduate Program at MIT
- ❖ Data from the US Census Bureau
- ❖ Advice to consider:

  "A Guide for potential grad students: why pursue a PhD?"
  https://www.gograd.org/graduate-school-guide-book/

Website to Apply to EECS

- https://apply.eecs.mit.edu/
- Website is active starting September 15th and will close December 1st.
- This website is specific for EECS only, and is NOT the general MIT admission website.
- Faculty begin reading applications in January.

Applying to Graduate School

THINKING THROUGH THE PURPOSE OF AN APPLICATION:

Reviewers want to know:

- Are you able to conduct independent research?

How is the research potential of an applicant determined?

- Undergraduate Subjects: selection & grades
- Statement of Purpose: career goals and aspirations
- Recommendation Letters

Undergraduate Preparation

For an undergraduate, a GPA of better than 3.5/4.0 is expected, but most admitted students have higher GPAs.
In EECS, at MIT, we consider closely:

- which technical subjects have been taken;
- grades in later semesters count more than early semesters;
- the GRE is **NOT** required for MIT EECS.

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Statement of Purpose

IMPORTANT: this is the only time that your voice is presented, so write this carefully!

Much more than a resume!

- Explain why you wish to attend graduate school.
- Why is EECS at MIT a good fit for you? Be specific!
- Describe your **research experience(s)**.
- Indicate research areas that you find exciting.

Do your homework by investigating the department website to identify research efforts and groups.

Letters of Recommendation

Work to obtain the best possible letters!

Strategy:

- Ask the letter writer if they are able to write the letter that you are seeking.
- If they cannot give you the letter you want, don't give them the information to submit the letter.
- Help the letter writer by providing your statement of purpose and a resume and any other material that will provide information about you.
- Give your letter writer plenty of notice before the deadline so they have the time and space to write you a thoughtful recommendation.

Letters of Recommendation, pt. 2

- Letters describing your contributions to a project are preferred to letters documenting your class participation.
- A letter **with detail** from someone that worked closely with you will be better than a vague letter from someone with an impressive title.
- The letter should help to convey your ability to carry out research, and ultimately independent research.

What NOT to do when crafting an application…

Refrain from providing a long list of activities…WHY?

- Research is a commitment to a single topic.
- List activities you enjoy, not numerous activities attempting to fill up your resume.
- Describe the few projects that were most successful, where you played a major role, or a project(s) that will be described by your letter writer(s).

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Apply to More than ONE School

- Each school may use different criteria for evaluation.
- An evaluator at one school may know your letter writers or your undergraduate program better.
- There are more applications than openings.
- Your research interests may not fit the graduate program.
- How many? Check to see if the programs you are applying to have application fee-waivers.

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Why I Chose MIT

A graduate of the Computer Science program, Sumaiya Nazeen is from Chandpur, Bangladesh. "MIT is every engineer's dream; when I was offered an opportunity to live out that dream and belong to a community of the greatest minds of today's world, I decided to take it." Dr. Sumaiya Nazeen '2019 is a postdoctoral research fellow at the Department of Biomedical Informatics at the Harvard Medical School and the Department of Neurology at the Brigham and Women's Hospital.

A graduate of the Electrical Engineering program, Sebastian Palacios is from Bogota, Columbia (and also lived in Miami, Florida). "I chose MIT because of its tradition of academic excellence, a problem-solving culture to address the world's great challenges, and a genuine caring community." Dr. Sebastian Palacios '2021 is a research scientist at MIT.

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Why I Chose MIT

A graduate of the Computer Science program, Danielle Olson is from Chantilly, Virginia. "I chose MIT for both undergraduate and graduate school because the people who come here are those who are determined to creating a brighter future, to use their education to hack the world and make it more like MIT. My most important priority has always been to connect my personal experiences with my education and work to make a positive social impact, and I believe that commitment is in MIT's DNA." Dr. Danielle Olson '2021 is a AI human factors research engineer at Apple.

Mengfei Wu is a graduate of the Electrical Engineering program from Singapore. "I chose MIT because during my visit to MIT, I really enjoyed talking to my research advisors and their group members." Dr. Mengfei Wu '2018 is a research scientist at the Institute of Materials Research and Engineering in Singapore.

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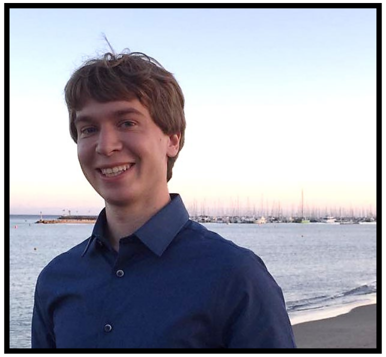
Why I Chose MIT



A graduate of the Electrical Engineering program, Greg Stein is from Ardsley, New York. "Seeking both personal and academic growth during graduate school, I was drawn to MIT's EECS Department, whose welcoming community of students and superb faculty have helped foster my graduate career." Dr. Greg Stein '2020 is Assistant Professor of CS at George Mason University.

Charles Mackin is a graduate of the Electrical Engineering program from Tucson, Arizona. "I decided to pursue my graduate studies at MIT for several reasons: 1) I felt that I was an excellent match for my principle investigator's lab both in terms of research interests and experience as well as personality, 2) the greater Boston area is a true intellectual and entrepreneurial hub with a ton to offer, and 3) because I always assumed that as an electrical engineer by training I would ultimately end up in Silicon Valley. I really wanted the opportunity to build an additional network in the Boston/Cambridge area (while earning a PhD from what is arguably the world's finest technology research institution) before moving out West to Silicon Valley." Dr. Charles Mackin '2018 is research scientist at IBM Research- Almaden.

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Graduate Programs in EECS at MIT

Master of Science Degree

All students must first earn a Master of Science degree

- Requires 4 graduate-level EECS subjects
- Requires completion of a thesis on research
- Complete the SM degree in two years

Note: if you enter with a Master of Science degree, you do not need to obtain the SM degree from EECS at MIT (but you may elect to do so).

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Doctor of Philosophy Degree

Program Requirements:

- Master of Science degree
- Successfully pass the technical qualification evaluation
- Successfully pass the research qualification examination
- One semester teaching assistantship
- Minor Program (2 subjects)
- Professional Perspective (2 units)
- Thesis with Public Defense

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Doctoral Qualification

- Technical Qualifying Evaluation (TQE):
  select four subjects from a grid of accepted subjects (earn grades of 3As, 1B)

- Research Qualifying Exam (RQE):
  prepare a technical paper and present research to a committee of two Profs- answer questions related to research endeavor carried out at MIT

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How to Pay for Graduate School?

Typically, your tuition is covered and you receive a monthly stipend*.

Types of support in EECS at MIT:

- Research Assistantships (RAs)
- Fellowships (both external and internal to MIT)
- Teaching Assistantships (TAs)

In EECS at MIT admission includes support for the first year (stipend, tuition & medical insurance); full support is typical for entire duration of graduate program.

*$4369 while earning SM degree
*$4695 while earning PhD degree
[stipend beginning Fall 2025 with annual increases]

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Graduate Student Association in EECS

Graduate Student Association Leadership Team for 2024

Graduate Women in EECS

Boston Harbor Sunset Cruise

MIT Program for Exemplary Mentoring

The departments participating in the MIT PEM (Program for Exemplary Mentoring) include: Electrical Engineering and Computer Science, Chemical Engineering, Mechanical Engineering, Health Sciences and Technology, Bioelectrical Engineering, Nuclear Science and Engineering and Civil and Environmental Engineering. The graduate students shown here are attending the PEM Summer Retreat in Falmouth, MA.

Data From the US Census Bureau

http://www.census.gov

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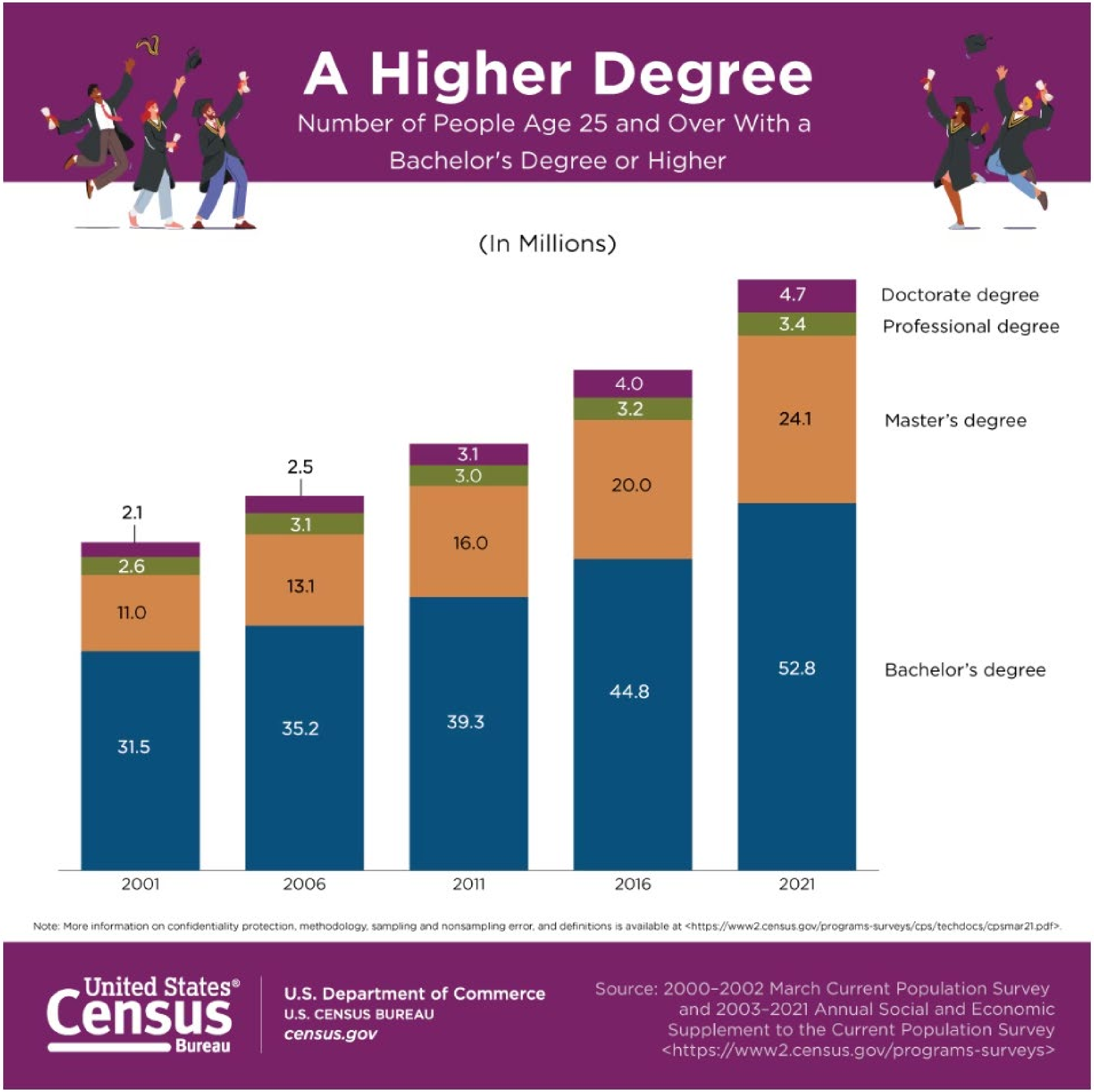

A Higher Degree
Number of People Age 25 and Over With a Bachelor's Degree or Higher
(In Millions)
Doctorate degree
Professional degree
Master's degree
Bachelor's degree
2001
2.1
2.6
11.0
31.5
2006
2.5
3.1
13.1
35.2
2011
3.1
3.0
16.0
39.3
2016
4.0
3.2
20.0
44.8
2021
4.7
3.4
24.1
52.8
Note: More information on confidentiality protection, methodology, sampling and nonsampling error, and definitions is available at <https://www2.census.gov/programs-surveys/cps/techdocs/cpsmar21.pdf>.
United States® Census Bureau
U.S. Department of Commerce
U.S. CENSUS BUREAU
census.gov
Source: 2000-2002 March Current Population Survey and 2003-2021 Annual Social and Economic Supplement to the Current Population Survey <https://www2.census.gov/programs-surveys>

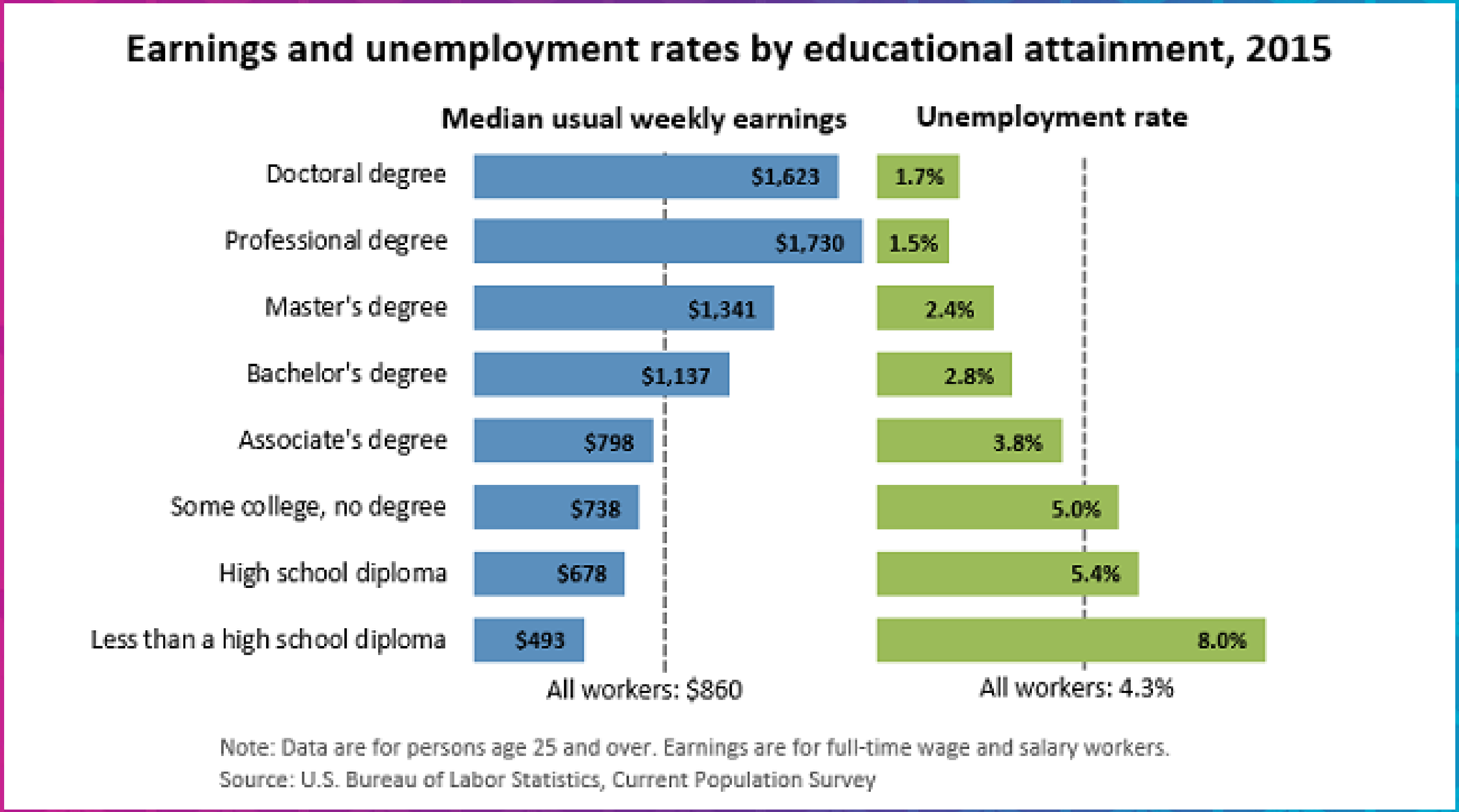
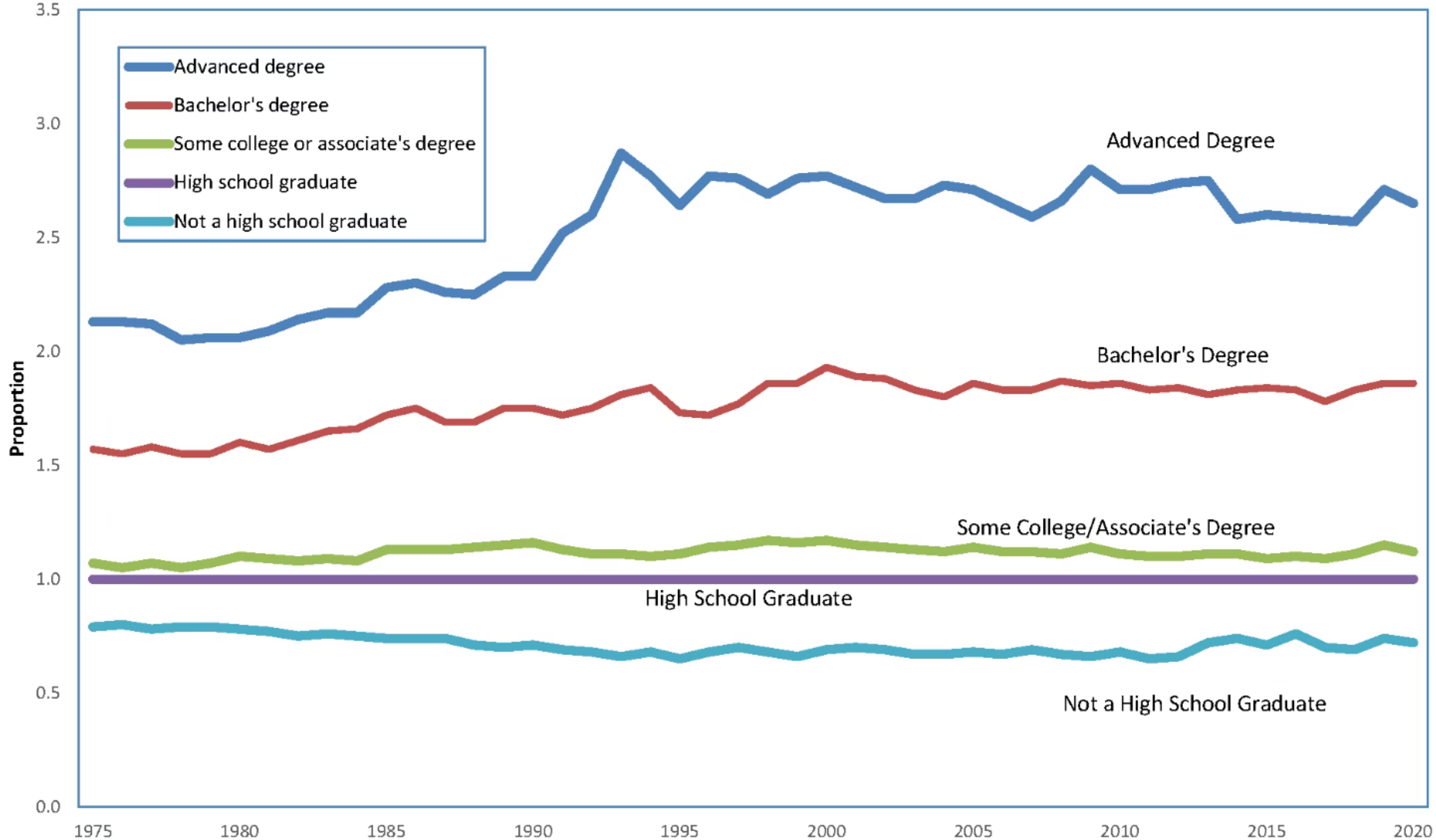

Earnings and unemployment rates by educational attainment, 2015
Median usual weekly earnings
Unemployment rate
Doctoral degree
$1,623
1.7%
Professional degree
$1,730
1.5%
Master's degree
$1,341
2.4%
Bachelor's degree
$1,137
2.8%
Associate's degree
$798
3.8%
Some college, no degree
$738
5.0%
High school diploma
$678
5.4%
Less than a high school diploma
$493
8.0%
All workers: $860
All workers: 4.3%
Note: Data are for persons age 25 and over. Earnings are for full-time wage and salary workers.
Source: U.S. Bureau of Labor Statistics, Current Population Survey

Figure 10: Average Earnings by Educational Attainment as a Proportion of the Average Earnings of High School Graduates: 1975-2020



Sources: U.S. Census Bureau. 1976-2002 March Current Population Survey, 2003-2021 Annual Social and Economic Supplement to the Current Population Survey. For more information on confidentiality protection, sampling error, nonsampling error, and definitions, see <https://www2.census.gov/programs-surveys/cps/techdocs/cpsmar21.pdf>

United States Census Bureau

Employment by Major Occupational Group

Data presented by major occupational group for 2012, projected to 2022, and median annual salary:
http://www.bls.gov/emp/ep_table_101.htm

Education by Detailed Occupation

Data presented for educational attainment by detailed occupation, 2012:
http://www.bls.gov/emp/ep_table_111.htm

Work-Life Earnings with Education

Educational Attainment	Work-Life Earnings	Margin of Error*
None to 8th grade	$936,000	$7,000
9th to 12th grade	$1,099,000	$7,000
High School graduate	$1,371,000	$3,000
Some college	$1,632,000	$5,000
Associate's degree	$1,813,000	$9,000
Bachelor's degree	$2,422,000	$8,000
Master's degree	$2,834,000	$13,000
Professional degree	$4,159,000	$33,000
Doctorate degree	$3,525,000	$29,000

Synthetic Work-Life Earnings by Educational Attainment
(In dollars. For information see:
https://www.census.gov/library/publications/2011/acs/acs-14.html)

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